TANMOY ROY *PhD*

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SUMMARY

AI and Machine Learning professional, consultant, and research guide with 19+ years spanning industry and academia. Currently a Senior Data Scientist leading **AI/ML** solutioning and team building. Work spans *speech processing, clinical NLP, computer vision, medical imaging, and IoT-edge computing*, with a current focus on on-prem SLMs that can be inspected and deferred rather than only made more fluent.

Doctoral research was **speech emotion recognition** (SER): a novel feature set (SEGRT) and a speech endpoint-detection algorithm, with peer-reviewed publications. Awarded the *Global Excellence Stature* (GES) scholarship at the **University of Johannesburg** (UJ) and completed the PhD in three years, then continued as a **post-doctoral research fellow** under the same fellowship.

Started at **SIEMENS** as a programmer on telecom, electronic-vehicle, and smart-grid billing systems. Built fluency in C, C++, Java, SQL and full project life cycles.

EXPERIENCE

DeliverHealth Solutions

Remote, India [July '24 – Present]

Senior Data Scientist

Lead AI/ML product ideation and organization-wide adoption. Design and deploy on-prem specialist models with product owners and partners including Google and Microsoft. Current bar: models small enough to inspect and honest enough to defer, not merely more fluent.

Medical Coding Co-pilot: SLM fine-tuned on proprietary historical data in a continuous learning loop to assist human coders and cut time-per-document. White-box logits used as a confidence meter; on this task a specialized on-prem SLM outperformed a generic LLM (Gemini) baseline, with right-vs-wrong AUROC moving from $\sim$0.60 to $\sim$0.90.

Automatic Medical Note Generation: Template-based physician note generation on a self-service platform, using Gemini 3.1 Pro and proprietary historical data.

RAG-based Medical Correction Engine: Retrieval-augmented correction of ASR drafts before delivery, reducing manual cleanup and lifting document quality.

Medical ASR (Med-ASR): In-house speech-to-text engine on proprietary data. Fine-tuned Gemini-1.5-Pro and Whisper; reached 12% medical WER without sending encounter audio to a public cloud.

Cloudcraftz Solutions

Kolkata, India [Nov '22 – Jun '24]

Senior Data Scientist

Led and mentored a team spanning data science, quantitative analytics, and software engineering, delivering finance-domain systems.

Option Trading Strategy Backtest: Production backtest for complex option strategies. Mix of software engineering, financial analytics, and data science.

Explainable AI: SHAP and accumulated local effect (ALE) visualizations for model transparency; built the product end to end.

Data Visualization Platform: Low-intervention analytics UI; led platform development.

Veersa Technologies

Kolkata, India [May '22 – Oct '22]

Data Scientist (Consulting)

Primary responsibility was ML model building with both tabular and image data. Other responsibilities involve data engineering, ML model productionizing, and ML team building. Executed projects in the healthcare domain.

Insurance Claim Denial Prediction: The primary target of this project is to predict whether claims submitted by the health institute will get denied or approved. Even as further analysis the reasons behind the denial are also depicted. Reducing the number of denials can have a great impact on the revenue of the health institute. Data is extracted in tabular form with relevant features and after required pre-processing that data is used as training data for the XGBoost classifier. Model accuracy was 89% although there was further scope for improvement.

Patient Data De-Duplication: Broken chain of information is a challenge for healthcare providers because this affects service quality and consumes storage space. The **record linkage** concept is used to solve this problem.

Disease Detection from X-Ray images: Detecting diseases from an X-ray can lead to a significant reduction of time and manual intervention. Used ResNet34 model to achieve 84% accuracy.

Clinical Trial Enrollment Prediction: This was a proof of concept on the publicly available high-quality data where the target was to predict the number of people finally enrolled for a clinical trial. The prediction accuracy achieved was 75% but there was further scope for improvement.

Handwritten Medical Form Reader: Digitization of handwritten medical forms is a big challenge for health service providers mainly due to varying hand-writing styles. AWS Textract is used to extract handwritten texts with high accuracy and the solution is built around the Textract output.

Research and Training Consultancy

Kolkata, India [Feb '20 – Present]

Research Consultant

Research consultancy across computer vision, NLP, and data science:

Human Fall Detection (CV): Fall detection from surveillance frames. Evaluated current deep-learning modules. Proposed a Masked R-CNN semantic segmentation model [2] at 94% F1 and 89% IoU; next module is human pose estimation.

Text Classification (NLP): Multi-head attention classifier [1] above 96% test accuracy, compared with Naive Bayes, logistic regression, random forest, and MLP.

Polyp Detection (Medical Imaging): U-Net architecture for polyps in colonoscopy images, aimed at cancerous-polyp detection.

Electronic Health Record (EHR) and Process Mining: 30-day readmission prediction from EHR. Process-mining features into sequence-to-sequence models; 88% test accuracy.

Face Mask Detection (CV): Mask detection from surveillance cameras during Covid-19 using MobileNet V2 and Inception ResNetV2.

IoT-Edge Computing: PhD research proposal for an edge IoT forensic model: online incident identification, evidence management, live threat detection, and forensic data reduction, primarily deep learning.

Training Consultant

Training tracks for students, career-switchers, and practising engineers:

Data Science and Machine Learning: Visualization, statistics, regression, classification, clustering, and time-series, with Python and R labs at audience-appropriate depth.

Deep Learning: ANN, CNN, RNN, through attention-based models.

Python for ML and DL: numpy, scipy, pandas, pytorch, tensorflow; OOP; Flask for serving models.

Practical Machine Learning (MLOps): Packaging, CI/CD, web serving, maintenance and reporting.

Cloud and Docker: AWS data toolchain (DMS, RDS, EMR, SageMaker) and ML/DL containers.

University of Johannesburg

Johannesburg, South Africa [Jul '16 – Sep '21]

Post Doctoral Research Fellow [Oct '19 – Sep '21]

Built a feed-forward network [3] on the doctoral SER feature set [8]. Wrote two surveys: emotion recognition in the 4th industrial revolution [5] and SER classification techniques [4]. Opened collaborations with King Saud University [2] and BITS Pilani [1].

Doctoral Research Fellow [Jul '16 – Jun '19]

Thesis: "Detecting emotions from speech using Machine Learning Techniques," completed in three years. Proposed SEGRT (*subjective emotional gap reduction technique*) [8, 6] and a speech endpoint-detection algorithm for pre-processing [7].

SIEMENS

Kolkata, India [Jan '07 – Sep '15]

Associate Consultant [Jan '13 – Sep '15]

• Designed and implemented a Telecom Service Provisioning system based on a RedHat Linux multi-tiered application using Java Spring Framework. Technologies used *Java Spring, Oracle*.
• Customization of Telecom Service Provisioning system using *Java, SOAP, XML, Oracle* on *Linux* platform.
• Designed and implemented CRM Service Provisioning integration module for GSM-R Billing system using *Java, Oracle PL/SQL*.
• Designed and implemented the Billing module for GSM-R Billing system using *C/C++, Oracle PL/SQL* on *Linux* platform.

Sr. System Engineer [Jan '11 – Dec '12]

• Designed and implemented a Charging module for the Electronic Car billing system using *C/C++ and Oracle*.
• Designed and implemented GUI based Telecom Call Analysis system. Features include Call tracking, Zone based call analysis with Graphical simulations. Used *C++ based Qt* GUI Library.

System Engineer [Jan '09 – Dec '10]

• Designed and implemented Bill Printing and Bill By Email module for a Telecom Billing system. **Technologies**: Java, ProC/C++, Oracle PL/SQL.
• Designed and implemented a Charging module for a Telecom Interconnect Billing system using Java and Oracle. **Technologies**: Java, Oracle.

Associate System Engineer [Jan '07 – Dec '08]

• Single handedly provided System Support for IVR and USSD Solution. Technologies used: *Java, C++, Linux, awk script, shell script*.
• Developed an automated monitoring system using *Java, shell script, awk script* to improve client experience.

RESEARCH EXPERTISE

- Drawn to research while at SIEMENS; several of those solutions remain in production.
- During the PhD, proposed a novel SER feature set [8, 6] and a speech endpoint-detection algorithm in a high-impact journal [7].
- Interdisciplinary ML/DL: computer vision [2], NLP [1], and medical imaging.
- Seven publications and one paper in preparation; open to further individual and collaborative work.

RESEARCH SUPERVISION & GUIDANCE

- **Guided a PhD scholar from King Saud University, Saudi Arabia (completed).**
  Area of Research: Computer Vision
  Research Title: Detection of human fall from a surveillance camera.
- **Supervised an MTech student from BITS Pilani, India (completed).**
  Area of Research: Computer Vision
  Research Title: Visual Scene Understanding for Autonomous Vehicles.

TEACHING EXPERTISE

Machine Learning and Data Science topics:

- Data scaling: normalize, standard scaler, one-hot.
- Data cleaning: missing value imputation, outlier detection.
- Feature selection: feature importance, correlation, Covariance, t-test, F-test, Chi^2 test, ANOVA.
- Feature extraction: PCA, LDA, ICA, SVD, NMF.
- Hypothesis testing, parametric and non-parametric tests.
- Regression: OLS, Ridge, Lasso, Elastic Net, Polynomial, Bayesian Ridge, SGD, Gaussian Process.
- Classification: Logistic Regression, KNN, Naïve Bayes, Decision Tree, Ensemble Methods- Bagging (Random Forest), Boosting (Ada Boost, Gradient Boosting, XGBoost).
- Clustering: KMeans, Hierarchical
- Time-series analysis

Deep Learning models: ANN, CNN, RNN, LSTM, Auto-Encoders, Attention

Programming: Python, C/C++, SQL, R, Java.

TECHNICAL EXPERTISE

- **Programming Languages:** Python, C/C++, R, Matlab, Java, JavaScript, SQL, Scala, Julia.
- **Libraries:** ollama, langchain, langgraph, unsloth, pytorch, tensorflow, pandas, matplotlib, scipy, opencv, pyspark.
- **Technologies/Platforms:** On-prem GPU inference, Cloud (AWS, GCP), Linux, Docker, MLOps (Kube-Flow), REST, SOAP, XML.
- **Databases:** RDBMS (Oracle, PostgreSQL, SQL Server, MySQL), NoSQL (MongoDB, DynamoDB, Cassandra), Cloud DB (Amazon RDS).

EDUCATION

Doctor of Philosophy

2019, University of Johannesburg — Johannesburg, South Africa
Thesis Title: Detecting Emotions From Speech Using Machine Learning Techniques.
Area of research: Applied Machine Learning and Deep Learning, Speech Signal Processing.

Master of Science

2013, Lovely Professional University — Kolkata, India
Specialization: Information Technology.
Subjects: Computer Organization, Programming(C++/Java), Data Structure,
Computer Graphics, Artificial Intelligence, System Software.

Post-Graduate Diploma in Computer Applications

2004, DOEACC Society, Ministry of IT, Govt. of India. — Kolkata, India
Advanced Diploma in Computer Applications (A-Level).

Bachelor of Science

2002, University of Calcutta — Kolkata, India
Economics(Hons), Computer Science, Mathematics

CERTIFICATES

1. **Deep Learning with PyTorch : Neural Style Transfer**
   Coursera
2. **Neural Network from Scratch in TensorFlow**
   Coursera
3. **Billing and Revenue Management**
   Oracle University
4. **Diploma in Computer Networking**
   CMC Limited

PUBLICATIONS

[1] Sinha J and Roy T. "Attention Based Encoder Architecture for Automatic Text Classification: A Case Study on Text-news Classification". In: *International Journal of Scientific Research in Engineering And Management* (2022). DOI: 10.55041/IJSREM12107.

[2] Alanazi T and Roy T. "Generating Human Segmentation Dynamically in Multi-camera Setup Using Deep Learning". In preparation. 2022.

[3] Roy T, Marwala T, and Chakraverty S. "Chapter 12 - Speech emotion recognition using deep learning". In: *New Paradigms in Computational Modeling and Its Applications*. Ed. by Snehashish Chakraverty. Academic Press, 2021, pp. 177–187. ISBN: 978-0-12-822133-4. DOI: https://doi.org/10.1016/B978-0-12-822133-4.00009-8.

[4] Roy T, Marwala T, and Chakraverty S. "A Survey of Classification Techniques in Speech Emotion Recognition". In: *Mathematical Methods in Interdisciplinary Sciences*. John Wiley & Sons, Ltd, 2020. Chap. 3, pp. 33–48. ISBN: 9781119585640. DOI: https://doi.org/10.1002/9781119585640.ch3. eprint: https://onlinelibrary.wiley.com/doi/pdf/10.1002/9781119585640.ch3.

[5] Roy T, Marwala T, and Chakraverty S. "Advancements and Role of Emotion Recognition in the 4th Industrial Revolution". In: *The Disruptive Fourth Industrial Revolution: Technology, Society and Beyond*. Ed. by Wesley Doorsamy, Babu Sena Paul, and Tshilidzi Marwala. Cham: Springer International Publishing, 2020, pp. 179–203. ISBN: 978-3-030-48230-5. DOI: 10.1007/978-3-030-48230-5_8.

[6] Roy T, Marwala T, and Chakraverty S. "Speech Emotion Recognition Using Neural Network and Wavelet Features". In: *Recent Trends in Wave Mechanics and Vibrations*. Ed. by S Chakraverty and Paritosh Biswas. Singapore: Springer Singapore, 2020, pp. 427–438. ISBN: 978-981-15-0287-3. DOI: 10.1007/978-981-15-0287-3_30.

[7] Roy T, Chakraverty S, and Marwala T. "Precise detection of speech endpoints dynamically: A wavelet convolution based approach". In: *Communications in Nonlinear Science and Numerical Simulation* 67 (2019), pp. 162–175. ISSN: 1007-5704. DOI: https://doi.org/10.1016/j.cnsns.2018.07.008.

[8] Roy T et al. "Introducing New Feature Set based on Wavelets for Speech Emotion Classification". In: *2018 IEEE Applied Signal Processing Conference (ASPCON)*. 2018, pp. 124–128. DOI: 10.1109/ASPCON.2018.8748666.

SCHOLARSHIPS

- Received Global Excellence Stature (GES) Doctoral fellowship at the University of Johannesburg in 2016.
- Received GES Post Doctoral Fellowship at the University of Johannesburg in 2019.

JOURNAL REVIEWER

- **International Journal of Systems Science.**
  Tailor and Francis

REFERENCES

1. **Lance Mansfield** (LinkedIn)
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2. **Nimish Sanghi** (LinkedIn)
   Founder at Cloudcraftz, SOAIS
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3. **Professor Snehashis Chakraverty** (LinkedIn)
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